# Second Partial

Fco. Vazquez

Tec de Monterrey

# *Class Activity*

Solve the following derivatives as preparation for the interpartial exam. Reduce the answer as much as possible.

$$f_1(x) = e^{\cos(x)} \qquad f_2(x) = \frac{3}{2+e^x} \qquad f_3(x) = e^{-x^4}\sin(x)$$

$$f_4(x) = \log_{10}\left[10^x + 1\right] \qquad f_5(x) = \left(\frac{3}{2}\right)^x \qquad f_6(x) = 2x \cdot e^{2/3x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}u(x)v(x) = u'(x)v(x) + u(x)v'(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\frac{u(x)}{v(x)} = \frac{1}{v^2(x)}\left[u'(x)v(x) - u(x)v'(x)\right]$$

$$\frac{\mathrm{d}}{\mathrm{d}x}u(v(x)) = u'(v(x))v'(x)$$

$$a^n \cdot a^m = a^{n+m}, \qquad \log_n[a] + \log_n[b] = \log_n[a \cdot b]$$

$$(a^n)^m = a^{n \cdot m}, \qquad a\log_n[b] = \log_n[b^a]$$

$$\frac{1}{a} = a^{-1}, \qquad \log_n[a] - \log_n[b] = \log_n\left[\frac{a}{b}\right]$$

$$a^{\log_a[x]} = x = \log_a[a^x], \qquad \log_n[x] = \frac{\log_m[x]}{\log_m[n]}$$